

Preference-Aware Decision-Making and Trajectory Planning: An Integrated Framework for Personalized Autonomous Driving

Abstract: Autonomous vehicles (AVs) have the potential to significantly enhance road safety, especially in scenarios such as following and lane-changing. However, most existing strategies neglect user-specific driving styles (e.g., aggressive, normal, or cautious), which limits their adaptability to diverse demands. To address this gap, we propose a personalized autonomous driving framework that integrates driver profiling, risk quantification, preference-aware decision-making, and trajectory planning. Specifically, a driving style analysis is first conducted, whereby deviations between naturalistic and intelligent driving are quantified with an intelligent driver model (IDM) and then three distinct driving style clusters are derived via a gaussian mixture model (GMM). These clustered styles provide parameterized input for the downstream decision-making. Building upon the driver profiling, a dual-dimensional (lateral-longitudinal) risk assessment model is introduced to evaluate real-time collision risk, including cut-in and cut-out maneuvers, thereby ensuring vehicle safety in dynamic traffic conditions. This risk evaluation is further incorporated into a finite state machine (FSM)-based decision-making framework, which combines risk levels with driver profiles to generate individualized strategies, such as longitudinal actions (accelerating, maintaining speed, or braking) and lateral maneuvers (car-following or lane-changing). Furthermore, the trajectory planning strategy is proposed through a dual-component design: a personalized adaptive cruise control (PACC) scheme for longitudinal motion, and a lane-changing planner that integrates fuzzy logic with quintic polynomials. The latter is enhanced with a speed pre-adjustment (SP-A) mechanism, allowing efficiency improvements without compromising user preferences. The proposed framework is evaluated in multi-vehicle interaction simulations on three-lane highways. Results confirm the proposed personalized model enables safe, comfortable, and user-tailored driving, while significantly outperforming traditional single-style approaches in terms of traffic efficiency.

Keywords: Autonomous vehicle, driving style, risk assessment, decision-making, trajectory planning

1. Introduction

The emergence of advanced driver-assistance systems (ADAS) and autonomous vehicles (AVs) has been viewed as a promising solution to reduce crashes caused by human error (Li et al., 2018; Khattak et al., 2020). AV development aims to minimize human intervention while ensuring safe, comfortable, and efficient driving. However, most existing strategies neglect both users driving styles and their acceptance of autonomous driving. Human drivers develop diverse driving styles over long periods of experience, and a uniform autonomous driving strategy may be difficult for them to accept. For instance, aggressive drivers tend to prioritize efficiency, whereas cautious drivers place greater emphasis on smoothness and stability. Therefore, exploring personalized autonomous driving strategies is critical to building user trust (Sun et al., 2020). Similarly, some researchers have also highlighted the importance of differentiating driving styles in autonomous driving scenarios (Lu et al., 2022; Nguyen et al., 2023). As a result, AVs should be equipped with advanced decision-making and trajectory planning systems capable of generating driving strategies and local trajectories that are not only safe and comfortable, but also aligned with individual driver or passenger preferences. This personalization is key to meeting diverse user needs and improving

the overall acceptance of autonomous driving technology.

An advanced personalized driving system typically consists of four key modules: driving style recognition, driving risk assessment, personalized decision-making, and trajectory planning. With the advancement of environmental perception, onboard sensors are able to detect surrounding vehicles, and risk assessment models can then be applied to evaluate potential collision risks. Meanwhile, driving styles can be classified from historical driving data, providing input parameters for both decision-making and trajectory generation. By integrating these components, the decision-making and trajectory planning modules jointly enable AVs and ADAS to produce behavioral strategies and motion trajectories that are adaptive to user preferences.

1.1. Driving style and characteristics

A personalized autonomous driving system should incorporate a dedicated module for driving style analysis, as driving styles reflect individual preferences and can be quantified through driving behavior metrics (Lajunen et al., 1997; Wang et al., 2017; Xue et al., 2019). Current research typically adopts data-driven approaches that integrate both subjective self-reports and objective driving data (Shirazi et al., 2017; He et al., 2025). Subjective assessments, often conducted via surveys or behavioral scales (Chen et al., 2023), are prone to individual bias, whereas objective assessments are generally regarded as more reliable. The latter generally relies on statistical and clustering methods applied to naturalistic or simulated driving data, where parameters such as speed, acceleration, and heading angle are analyzed to differentiate driving styles and characterize their behavioral patterns (Zhu et al., 2019; Li et al., 2019; Wang et al., 2019; Chen et al., 2023). Driving styles are commonly categorized into aggressive, normal (or moderate), and cautious, corresponding to high, medium, and low levels of risk acceptance, respectively (Xue et al., 2019; Ma et al., 2021). Clustering methods are frequently employed for the classification, as they do not require labeled data. However, their effectiveness depends strongly on the separability of the underlying data. Moreover, conventional clustering methods often fail to incorporate domain knowledge (e.g., vehicle dynamics, motion constraints), which may reduce their ability to capture realistic driving styles (Chen et al., 2023). The results are also sensitive to data imbalance or the presence of extreme samples, both of which can significantly undermine classification accuracy.

1.2. Driving risk assessment

Existing studies generally classify risk assessment methods into deterministic and probabilistic approaches. Deterministic methods rely on predefined rules to determine whether a collision is likely to occur (Chen et al., 2024; Vigne et al., 2024). They often use time-based, distance-based, or acceleration-based indicators such as Time to Collision (TTC), Time Headway (THW), Post-Encroachment Time (PET), and Minimum Safety Distance (MSD). Their main advantages lie in real-time performance, low computational demand, and strong interpretability (Anis et al., 2025). In contrast, probabilistic approaches describe the likelihood of collisions by modeling vehicle interactions in terms of spatial-temporal relationships. However, most existing methods remain longitudinally oriented, with limited capability in assessing lateral risks. One-dimensional indicators like TTC, THW, and MSD are insufficient for complex lateral scenarios. To overcome this limitation, some researchers have introduced the artificial potential field (APF) theory to represent dual-dimensional driving risks (Li et al., 2024). While APF-based methods capture both longitudinal and lateral interactions, they often oversimplify vehicle dynamics and environmental conditions, leading to reduced accuracy in highly dynamic traffic. Moreover, their real-time performance is limited, and parameter tuning (e.g., potential field shapes, attractive/repulsive weights) is highly subjective (Hu

et al., 2025). Another distinct limitation of traditional methods is their reliance on fixed safety thresholds, which neglect the variability in drivers' risk tolerance. As a result, these approaches often enforce overly cautious strategies, making them unsuitable for accommodating diverse driving styles such as normal or aggressive behaviors.

1.3. Vehicle decision-making

There are two mainstream approaches to vehicle decision-making: rule-based methods and learning-based methods. Rule-based methods are widely adopted in autonomous driving due to their ease of implementation, stability, and ability to generate real-time responses. Among them, the Finite State Machine (FSM) is one of the most mature and commonly used frameworks. In FSM-based models, the AV generates decision actions based on its current state, the surrounding traffic environment, and traffic rules. Transitions between driving states (e.g., lane-keeping, car-following, or lane-changing) are determined by factors like vehicle position, speed, and risk. This interpretable structure not only ensures transparency but also aligns well with human drivers' cognitive processes, making it especially suitable for incorporating driver-specific styles. For example, Vasic et al. (2016) developed an FSM-based lane-change model guided by risk assessment, while Nguyen et al. (2023) extended FSMs for collision avoidance by considering both surrounding vehicles and static obstacles. However, they generally overlook individual differences in driving style, which limits their ability to provide personalized decision-making. In contrast, learning-based methods such as imitation learning (Kober and Peters, 2010), reinforcement learning (He et al., 2022), and deep reinforcement learning (Kiran et al., 2022; Ye et al., 2019) have been explored for decision-making under uncertainty. While such approaches hold promise for end-to-end decision-making, they often require extensive training data, high computational cost, and carefully designed reward functions. Moreover, their performance in rare or highly complex scenarios remains unreliable (Vinge et al., 2024).

1.4. Vehicle trajectory planning

Trajectory planning translates vehicle decisions into vehicle trajectories, such as speed and acceleration/deceleration profiles. Trajectory planning methods are generally divided into global planning and local planning. Global planning methods such as Dijkstra, A*, D*, and rapidly-exploring random trees (RRT) are optimization-based and focus on finding minimum-cost paths. These methods require substantial computational resources and often produce trajectories with discontinuous curvature, which makes them more suitable for coarse-grained planning problems such as robotic pathfinding or route navigation. In contrast, vehicle lane-changing maneuvers demand fine-grained and smooth trajectories, for which parametric curves are commonly employed (Li et al., 2020). Examples include clothoid curves (Mouhagir et al., 2020), Bézier curves (Lattarulo et al., 2017), polynomial curves (Luo et al., 2016), and S-curves (Vinge et al., 2024). Clothoid curves ensure curvature continuity but involve computationally intensive integral operations and may still lack smoothness. Bézier curves are efficient but highly sensitive to control points, making localized adjustments challenging and limiting their applicability for personalized planning. Polynomial curves, by contrast, offer a favorable balance between computational efficiency, curvature continuity, and trajectory smoothness. They also allow straightforward adjustment of parameters such as speed, acceleration, and time, making them well-suited for incorporating driver or passenger preferences. Despite these advances, existing studies still fall short in differentiated trajectory planning that reflects individual driving styles. Additionally, most research on lane-changing neglects speed pre-adjustment in the target lane. Without this harmonization, the ego vehicle may enter the target lane

at a speed significantly different from surrounding traffic, undermining both safety and efficiency. Furthermore, prior work often isolates the lane-change maneuver itself while ignoring the continuity of pre-lane-change, in-lane-change, and post-lane-change phases, leading to fragmented motion and reduced behavioral consistency.

1.5. Contributions

This study proposes a holistic framework of personalized autonomous driving strategy that integrates individual driving preferences and validates its effectiveness using naturalistic highway driving data. The framework consists of personalized lane-change decision-making and trajectory planning modules, designed to accommodate diverse user driving styles. More specifically, driving styles and behaviors are characterized through naturalistic trajectory data, which serve as the basis for model development. Then, risk assessment, decision-making, and trajectory planning models are constructed to collaboratively control vehicle behaviors, with the goal of achieving safe, efficient, and personalized driving. The proposed approach is evaluated in highway scenarios involving interactions with multiple vehicles. The main contributions of this study are as follows:

(1) We propose a hybrid clustering method that combines data-driven and model-based elements to classify driving styles from naturalistic vehicle trajectory data. This approach not only identifies key behavioral parameters for each driving style but also mitigates the impact of imbalanced samples and extreme cases, ensuring robust parameter estimation.

(2) A novel 2D risk assessment framework is developed to evaluate risks in both the current and adjacent lanes throughout lane keeping, lane departure, and lane merging. By employing a dual-dimensional vector TTC metric, the module integrates driving safety and individual preferences, enabling more comprehensive and adaptive risk evaluation.

(3) We design a personalized decision-making module based on the FSM, where the state transition rules are adapted using the identified behavioral parameters of different driving styles. This allows the decision-making process to explicitly capture individual differences, leading to safer, more efficient, and more style-consistent driving strategies.

(4) We develop a trajectory planning module that integrates both lane-changing and lane-keeping maneuvers. During lane changing, a quintic polynomial fitted with a speed pre-adjustment (SP-A) strategy generates smooth and adjustable paths, while for lane keeping, a personalized adaptive cruise control (PACC) module ensures stability and comfort. By embedding driving style parameters, these modules enable the vehicle to generate safe, smooth, and personalized trajectories.

1.6. Paper organization

The remainder of this paper is organized as follows: Section 2 describes the classification of driving styles and the parameter identification process. Section 3 presents the details of the personalized driving decision-making based on risk assessment. Section 4 outlines the personalized vehicle trajectory generation method, including the personalized adaptive cruise control and lane change speed preconditioning. In Section 5, various driving scenarios are designed to validate the effectiveness of the proposed approach. Finally, Section 6 summarizes the conclusions and presents potential future research perspectives.

2. The overall framework

2.1. The personalized driving framework

In this study, we propose a personalized autonomous driving framework that integrates driver

preferences into risk assessment, decision-making, and trajectory planning, as shown in Fig. 1. A driver behavior profiling module is first introduced, where deviations between naturalistic data and IDM outputs are clustered into different driving styles using a GMM. These styles are characterized by metrics such as THW, TTC, and lane change duration (LCD), and the behavioral parameters are identified. A 2D risk assessment module then evaluates longitudinal and lateral collision risks in real time, producing personalized risk levels that guide decision-making. Based on an FSM structure, the decision-making module integrates these risk levels with driving styles to derive strategies such as car-following, lane-keeping, or lane-changing. These strategies are then passed to the trajectory planning layer. Finally, the trajectory planning module translates decisions into vehicle trajectories. It consists of two components: a PACC module and a lane-changing planner with a SP-A mechanism. The PACC module handles car-following driving by dynamically adjusting longitudinal speed based on real-time risk levels and driver preferences. The lane-changing planner combines SP-A with QPs to generate smooth, safe, and preference-aware trajectories for lane-change maneuvers.

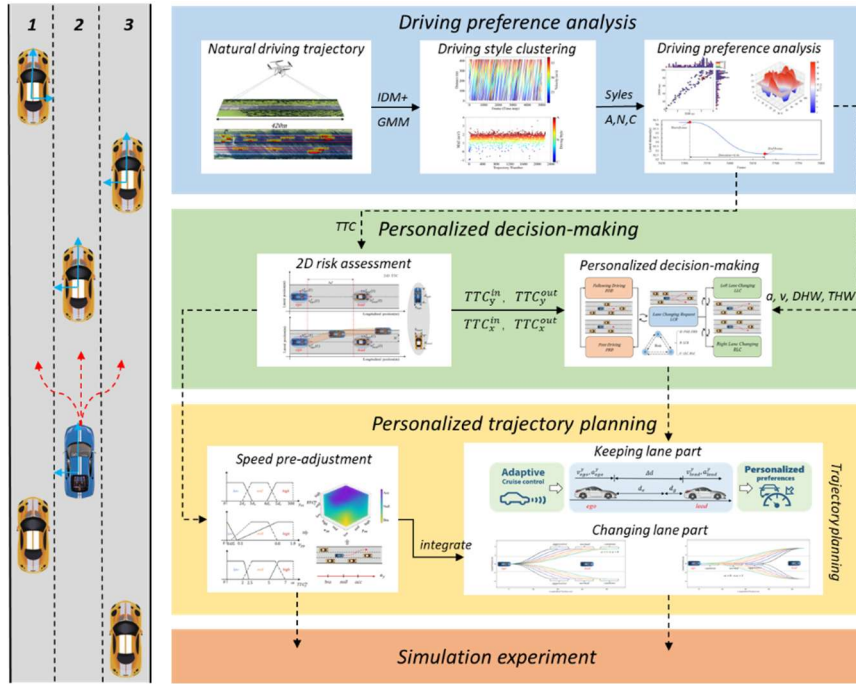


Fig. 1. The personalized driving framework.

2.2. Preliminary work

To design a personalized driving system that accurately reflects individual preferences, it is essential to identify the driving style parameters. Therefore, we first extract behavioral patterns from large-scale naturalistic trajectory data. In this study, we adopt the HighD dataset (<http://www.highD-dataset.com>), using drones to record highway traffic (Krajewski et al., 2018). It contains detailed information on vehicle types, dimensions, positions, speeds, and trajectories (as shown in Fig. 2), with over 110,000 vehicles and more than 45,000 km of coverage. The dataset's diversity, large scale, and high accuracy make it particularly well-suited for modeling and identifying driving preferences.

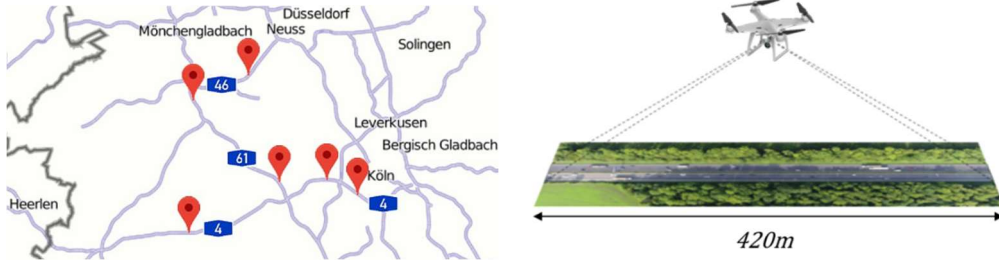


Fig. 2. Trajectory data collection.

First, we exclude trajectories involving severe speeding to ensure safety and compliance. Additionally, since the dataset may include unsafe trajectories caused by driver errors or data noise, we apply a minimum safe distance model in Eq. (1) to filter them out. This model ensures that even if the lead vehicle suddenly performs emergency braking, the ego vehicle can still respond in time to avoid collisions, thereby preventing unsafe samples from biasing the personalized driving system.

$$\begin{cases} d_{safe} = d_r + d_b + d_s - d_l \\ d_b = \frac{v_{ego}^2 - v_{lead}^2}{2a_{dmax}} \\ d_r = v_{ego} t_r \end{cases} \quad (1)$$

where d_{safe} represents the minimum safe distance that should be maintained between the ego vehicle and the lead vehicle. d_r is the distance traveled by the ego vehicle during the driver's reaction time and the vehicle's braking system response time. d_b is the distance the ego vehicle travels while braking at maximum deceleration until its speed matches that of the lead vehicle. d_0 is the minimum distance that should be maintained when both vehicles are at a complete stop. d_l is the distance traveled by the lead vehicle during the ego vehicle's braking process. a_{dmax} denotes the maximum deceleration of the ego vehicle. t_r is the total response time, including both the driver's reaction time and the braking system's response time.

Naturalistic datasets often show imbalanced distributions of driving styles, which can cause clustering results to deviate from reality (Jing et al., 2025; Ma et al., 2023; Xing et al., 2025). To address this, we propose a hybrid clustering method that combines data-driven analysis with model-based domain knowledge, integrating naturalistic driving datasets with the IDM to make the classification more robust. Specifically, we compute the expected acceleration from IDM under given driving conditions and compare it with the observed acceleration in the dataset. The error between actual and IDM-predicted acceleration reflects the driver's behavioral tendency. Since the IDM represents a conservative, safety-oriented style, it serves as the baseline. A larger mean acceleration error (MAE) indicates a more aggressive driving style. The MAE is then used as the clustering metric, as defined in Eq. (2).

$$MAE = \frac{\sum_{t=0}^T |a_A(t) - a_P(t)|}{T} \quad (2)$$

where $a_A(t)$ denotes the actual acceleration in natural driving, $a_P(t)$ represents the acceleration predicted by the IDM, and T is the trajectory period.

Next, a GMM is applied to cluster the vehicles based on their average acceleration deviation. The GMM is used to model the data as a mixture of multiple Gaussian distributions, enabling more flexible clustering than hard-assignment methods (Vlassis and Likas, 2002; Zhao et al., 2020; Shi et al., 2025). The mathematical formulation of the GMM is shown in Eq. (3).

$$\begin{aligned}
P(x) &= \sum_{k=1}^K \pi_k N(X|\mu_k, \Sigma_k) \\
N(X|\mu_k, \Sigma_k) &= \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{d/2}} \exp\left(-\frac{1}{2}(X - \mu_k)^T \Sigma_k^{-1} (X - \mu_k)\right)
\end{aligned} \tag{3}$$

where π_k is the weight of the k -th Gaussian component. $N(X|\mu_k, \Sigma_k)$ denotes the Gaussian distribution with mean μ_k and covariance matrix Σ_k . d is the clustering feature dimension of the trajectory dataset. We further analyze the behavioral characteristics associated with different driving styles. To meet the parameter requirements of the personalized autonomous driving system, we focus on key behavioral indicators such as acceleration, DHW, THW, TTC, and LCD. These indicators provide essential parameter inputs for the personalized decision-making and trajectory planning modules.

3. Personalized decision-making based on driver preferences

In conventional autonomous driving systems, the decision-making module usually evaluates driving risk based on the ego vehicle's state and surrounding environment, and then generates a safe driving plan (Hao et al., 2016; Yun et al., 2017; Zhang et al., 2025). However, such systems may not align with the preferences of different drivers or passengers. In this study, we design a personalized decision-making module that considers both driving risk and user preferences. Risk is assessed using a multi-perspective evaluation model that accounts for potential collisions in lane-keeping and lane-changing scenarios, serving as the basis for decision-making. To embed user preferences, driving style-specific behavioral parameters are incorporated into the decision rules, enabling an interpretable and preference-aware framework. The module is implemented as an FSM-based state transition model, which integrates vehicle state, collision risk, and driver preferences to generate decisions that are not only safe but also consistent with individual driving styles.

3.1. Multi-perspective risk assessment for vehicles

To accurately evaluate the dynamic risks during both lane-keeping and lane-changing maneuvers, we propose a dual-dimensional risk assessment framework that comprehensively considers threats from surrounding traffic participants. The risk assessment is divided into three phases: lane-keeping, lane cut-out, and lane cut-in. During lane-keeping, we use TTC to evaluate the collision risk between the ego vehicle and both the preceding and following vehicles. In lane-change scenarios, the ego vehicle faces not only longitudinal but also lateral collision risks. Therefore, for lane cut-out and cut-in phases, we develop a 2D vector TTC model to assess collision risks in both longitudinal and lateral directions, as illustrated in Fig. 3.

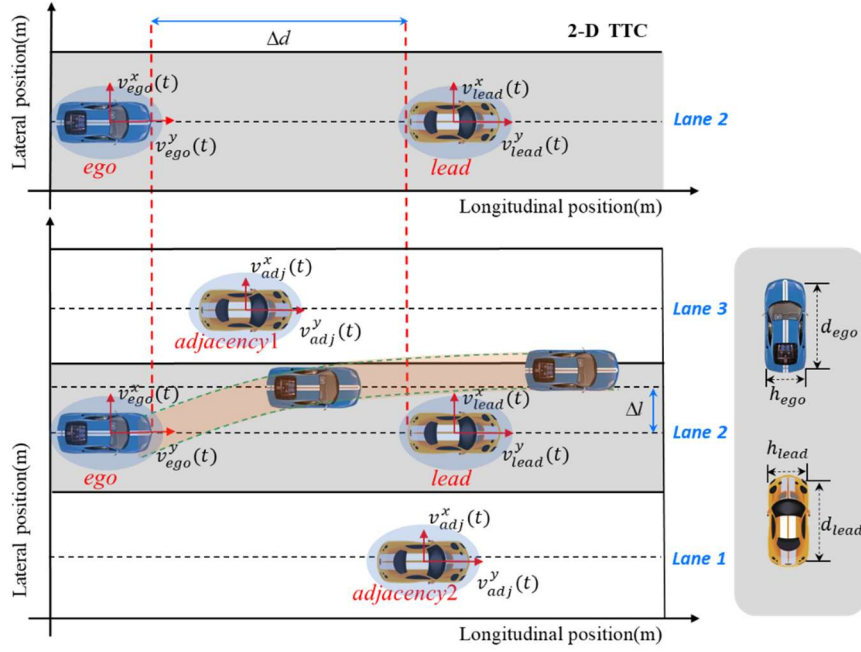


Fig. 3. 2D TTC during lane-changing.

As shown in Fig. 3, the diagram illustrates the calculation of 2D collision risk. The upper part represents longitudinal risk (TTC_y), determined by the relative distance and speed between the ego vehicle and the lead vehicle. The lower part shows lateral risk (TTE_x), which depends on whether the ego vehicle's lateral offset during longitudinal movement is sufficient to avoid overlap with surrounding vehicles. A greater lateral distance corresponds to a lower collision risk. Based on this, the collision risk of the entire driving process can be modeled. In lane-keeping, only longitudinal risk is considered. During lane-changing, the process is divided into two phases: *cut-out* and *cut-in*. The cut-out phase mainly involves interactions between the ego vehicle and the lead vehicle, while the cut-in phase is more complex, as risks arise from interactions with both the lead and adjacent vehicles in the target lane.

First, the collision risk in the cut-out scenario is modeled. The formula for calculating longitudinal collision risk in this context is given by Eq. (4) and Eq. (5).

$$TTC_y^{out} = TTC_y = S_t \frac{y_{lead}(t) - y_{ego}(t)}{v_{ego}^y(t) - v_{lead}^y(t)} \quad (4)$$

$$\begin{cases} y_{lead}(t) = y_{lead}^c(t) - \frac{1}{2}d_{lead} \\ y_{ego}(t) = y_{ego}^c(t) + \frac{1}{2}d_{ego} \end{cases} \quad (5)$$

where TTC_y represents the collision risk between the ego vehicle and the lead vehicle. $y_{lead}(t)$ and $y_{ego}(t)$ are the longitudinal positions of the ego vehicle and the lead vehicle at time t , respectively. $v_{lead}^y(t)$ and $v_{ego}^y(t)$ are their respective longitudinal speeds at time t . S_t is a constant that accounts for environmental conditions such as weather and road surface. Its value ranges from 0 to 1, with lower values indicating more hazardous conditions. $y_{ego}^c(t)$ and $y_{lead}^c(t)$ represent the longitudinal positions of the centers of the ego vehicle and the lead vehicle at time t , respectively. d_{lead} and d_{ego} are the vehicle lengths of the lead vehicle and the ego vehicle, respectively. The formula for calculating lateral collision risk in this context is given by Eq. (6) and (7).

$$TTC_x^{out} = TTE_x^{out} = \frac{\Delta l - (x_{ego}^c(t) - x_{lead}^c(t))}{v_{ego}^x(t) - v_{lead}^x(t)} \quad (6)$$

$$\Delta l = \frac{h_{ego} + h_{lead}}{2} \quad (7)$$

where TTE_x^{out} represents the lateral collision risk, defined as the time required for the ego vehicle to avoid a lateral collision. Δl is half the sum of the widths of the ego vehicle and the lead vehicle. $x_{ego}^c(t)$ and $x_{lead}^c(t)$ denote the lateral positions of the ego vehicle and the lead vehicle at time t , respectively. $v_{ego}^x(t)$ and $v_{lead}^x(t)$ represent the lateral velocities of the ego vehicle and the lead vehicle at time t , respectively.

During a lane change, the ego vehicle interacts not only with vehicles in the current lane but also with those in the target lane. To capture these interactions, a dedicated collision risk model is required for the cut-in phase. Based on the relative positions and speeds of the ego and adjacent vehicles, two cases can be distinguished: (1) No collision risk, when the adjacent vehicle is behind the ego vehicle and slower, or ahead of the ego vehicle and faster. (2) Potential collision risk, when the adjacent vehicle is behind but faster, or ahead but slower. In case (1), maintaining a minimum safe distance is sufficient. In case (2), however, a collision risk may arise during the cut-in process, requiring further evaluation (as shown in Fig. 4). To ensure consistent modeling across these cases, we extend the concept of TTC. If the adjacent vehicle is ahead, TTC is calculated directly. If it is behind, TTC is computed by treating the ego vehicle as the lead vehicle. To unify the approach, the absolute value of $\overrightarrow{TTC}_y^{in}$ is used, allowing the model to handle both front-to-rear relationships consistently. Furthermore, we interpret $\overrightarrow{TTC}_y^{in}$ directionally: $\overrightarrow{TTC}_y^{in} < 0$ indicates no collision risk (ego ahead and faster, or behind and slower), while $\overrightarrow{TTC}_y^{in} > 0$ indicates potential collision risk (ego ahead but slower, or behind and faster). By incorporating this directional $\overrightarrow{TTC}_y^{in}$ into the personalized decision-making module, the system obtains a more nuanced and reliable risk indicator, improving responsiveness and safety in cut-in scenarios.

$$TTC_y^{in} = \left| \overrightarrow{TTC}_y^{in} \right|, \overrightarrow{TTC}_y^{in} = S_t \frac{y_{adj}(t) - y_{ego}(t)}{v_{ego}^y(t) - v_{adj}^y(t)} \quad (8)$$

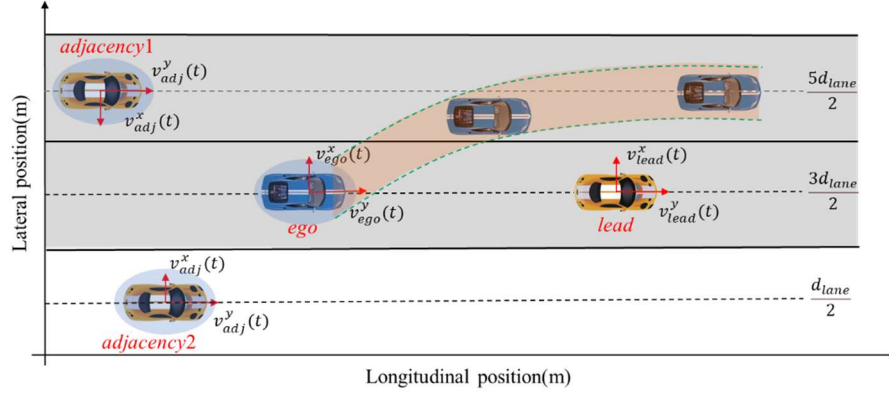
$$y_{adj}(t) = \begin{cases} y_{adj}^c(t) + \frac{1}{2}d_{adj}, y_{adj}^c \leq y_{ego}^c(t) \\ y_{adj}^c(t) - \frac{1}{2}d_{adj}, y_{adj}^c > y_{ego}^c(t) \end{cases} \quad y_{ego}(t) = \begin{cases} y_{ego}^c(t) - \frac{1}{2}d_{ego}, y_{adj}^c \leq y_{ego}^c(t) \\ y_{ego}^c(t) + \frac{1}{2}d_{ego}, y_{adj}^c > y_{ego}^c(t) \end{cases} \quad (9)$$

where TTC_y^{in} represents the longitudinal collision risk between the adjacent vehicle and the ego vehicle during a lane-change cut-in scenario. $y_{adj}(t)$ and $y_{ego}(t)$ denote the longitudinal positions of the adjacent vehicle and the ego vehicle at time t , respectively. $y_{adj}^c(t)$ and $y_{ego}^c(t)$ are the center positions of the adjacent vehicle and the ego vehicle at time t . $v_{adj}^y(t)$ and $v_{ego}^y(t)$ are the longitudinal speeds of the adjacent vehicle and the ego vehicle at time t . d_{adj} and d_{ego} represent the lengths of the adjacent vehicle and the ego vehicle, respectively. Similarly, the lateral collision risk during the cut-in scenario can be modeled as follows:

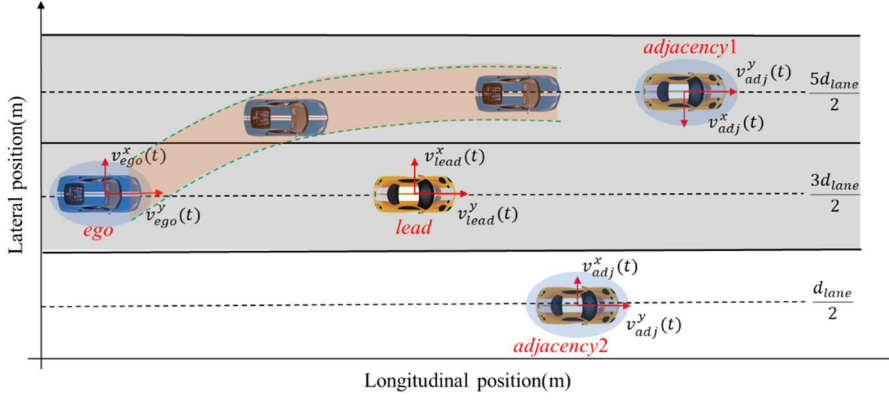
$$TTC_x^{in} = TTE_x^{in} = \frac{|x_{adj}^c(t) - x_{ego}^c(t)| - \Delta l'}{|v_{ego}^x(t)| + |v_{adj}^x(t)|} \quad (10)$$

$$\Delta l' = \frac{h_{ego} + h_{adj}}{2} \quad (11)$$

where, TTE_x^{in} represents the lateral collision risk between the ego vehicle and the adjacent vehicle during a lane-change cut-in scenario. $x_{adj}^c(t)$ and $x_{ego}^c(t)$ are the lateral positions of the adjacent vehicle and the ego vehicle at time t , respectively, while $v_{adj}^x(t)$ and $v_{ego}^x(t)$ are their respective lateral velocities at the same time.



a. Longitudinal position of the vehicle in the target lane behind the ego vehicle



b. Longitudinal position of the vehicle in the target lane ahead of the ego vehicle

Fig. 4. Illustration of collision risk of case (2) (example of left lane-changing).

Based on the above models, the driving risks associated with lane-keeping, cut-out, and cut-in scenarios have been quantitatively assessed. These assessments are then used to further determine whether the vehicle is exposed to potential risks. The core decision logic is embedded into the personalized decision-making module using an FSM, which is modeled in detail in Section 3.2.

3.2. Personalized decision-making framework based on FSM

FSM-based approaches emulate human driving behavior by embedding traffic knowledge and physical constraints into interpretable decision rules. In this study, the lane-change decision module adopts an FSM framework, as illustrated in Fig. 5. The model defines five driving states: *Free Driving (FRD)*, *Following Driving (FOD)*, *Lane Change Request (LCR)*, *Left Lane Changing (LLC)*, and *Right Lane Changing (RLC)*. The introduction of the LCR state ensures that a lane change is initiated only when the cut-out safety conditions are satisfied. This design prevents premature lane changes and subsequent forced returns to the original lane due to unmet safety requirements. Transitions between states are governed by the ego vehicle's position, speed, and risk level, as well as the surrounding traffic conditions.

FRD state. When there is no lead vehicle ahead in the ego vehicle's current lane, the vehicle enters the *FRD* state. In this state, the primary task is to adjust the ego vehicle's cruising speed to achieve smooth and comfortable driving. This is done by tracking the ego vehicle's real-time speed $v_{ego}^y(t)$ and the recommended speed for the current lane v_o . To accomplish this, the system calculates the speed difference $\Delta v = v_o - v_{ego}^y(t)$ and applies personalized acceleration based on the driving style, while staying within the acceleration constraints.

FOD state. When there is a lead vehicle ahead in the ego vehicle's current lane and the conditions in the adjacent lanes do not allow for a safe lane change, the ego vehicle enters the *FOD* state. In

this state, the vehicle performs personalized adaptive cruise control by matching the speed of the lead vehicle, while satisfying safety constraints.

$$TTC_y(t) \geq TTC_y^{p(A,N,C)} \geq TTC_{safe} \quad (12)$$

where, $TTC_y(t)$ represents the longitudinal collision risk between the ego vehicle and the lead vehicle at time t . $TTC_y^{p(A,N,C)}$ is the risk acceptance parameter corresponding to different driving preferences, determined through driver preference analysis. TTC_{safe} is the longitudinal collision risk threshold, representing the minimum safety constraint that must not be violated, regardless of the driving style.

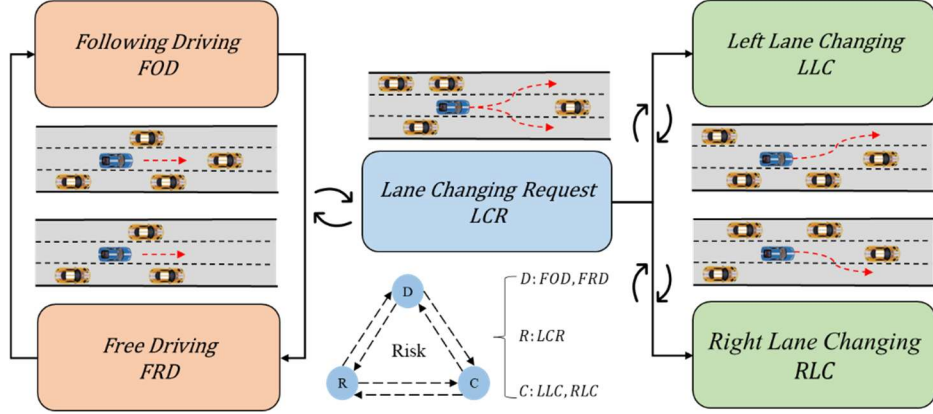


Fig. 5. FSM based decision-making model for vehicle.

LCR state. When the ego vehicle is in either the *FRD* or *FOD* state, the *LCR* state is simultaneously activated. In this state, the system evaluates traffic conditions in the adjacent lanes and estimates collision risks before, during, and after a potential lane change. A lane change is initiated only if the assessed risks satisfy both the minimum safety threshold and the risk tolerance associated with the current driving style. Specifically, a lane change request is triggered when:

$$f \begin{cases} TTC_y^{out}(t) \geq TTC_y^{p(A,N,C)} \geq TTC_{safe} \\ TTC_y^{out}(t) \geq TTE_x^{out}(t) \end{cases} \quad (13)$$

This condition indicates that at the current time t , the ego vehicle's longitudinal collision risk during a cut-out maneuver meets both the minimum safety threshold and the safety threshold specific to the given driving style. Additionally, the longitudinal collision time is longer than the lateral escape time. This ensures that the ego vehicle can safely and personally initiate a cut-out maneuver.

LLC state. If the left lane satisfies the lane-change requirements, its associated risk remains below the safety threshold, and the collision risk on the left is lower than that on the right, the ego vehicle transitions into the *LLC* state. The primary task is to leave the current lane and merge into the left lane.

RLC state. If the right lane satisfies the lane-change requirements, its associated risk remains below the safety threshold, and the collision risk on the right is lower than that on the left, the ego vehicle enters the *RLC* state. The objective is to exit the current lane and merge into the right lane. The criteria for assessing cut-in safety are analogous to those used for cut-out evaluation, and can be expressed by the following formula:

$$f \begin{cases} TTC_y^{in}(t) \geq TTC_y^{p(A,N,C)} \geq TTC_{safe} \\ TTC_y^{in}(t) \geq TTE_x^{in}(t) \end{cases} \quad (14)$$

The lane-changing conditions are further refined to determine which lane the ego vehicle should merge into at the time t . This decision is categorized into two cases, as described below:

$$\overrightarrow{TTC}_y^{in} < 0, \begin{cases} v_{(adj, left)}^y < v_{ego}^y \\ y_{(adj, left)} < y_{ego} \end{cases} \wedge \begin{cases} v_{(adj, right)}^y < v_{ego}^y \\ y_{(adj, right)} < y_{ego} \end{cases} \Rightarrow (X_{left}, X_{right}) = (1, 0) \quad (15)$$

$$\overrightarrow{TTC}_y^{in} \geq 0, \overrightarrow{TTC}_{(y, left)}^{in} \geq \overrightarrow{TTC}_{(y, right)}^{in} \geq 0, \text{ and } \overrightarrow{TTC}_y^{in}(t) \neq +\infty \Rightarrow (X_{left}, X_{right}) = (1, 0) \quad (16)$$

Using a binary probability function with values in (0, 1), the lane change direction transition condition can be simplified as follows:

$$X(t) = \begin{cases} 1, 0 < \overrightarrow{TTC}_y^{in}(t) \neq +\infty \text{ and } \overrightarrow{TTC}_{(y, right)}^{in}(t) > \overrightarrow{TTC}_{(y, left)}^{in}(t) \\ 0, \text{ otherwise} \end{cases} \quad (17)$$

where, $X(t)$ represents the lane change decision variable, $X(t)=1$ indicates an *RLC*, and $X(t)=0$ indicates an *LLC*. When the $\overrightarrow{TTC}_y^{in}(t)$ is greater than 0 and not equal to $+\infty$, and the $\overrightarrow{TTC}_{(y, right)}^{in}(t)$ is greater than that $\overrightarrow{TTC}_{(y, left)}^{in}(t)$, the ego vehicle will choose to cut-in the right lane. Otherwise, it will prioritize cut-in the left lane. The flow of vehicle decision rules and state transfer conditions is shown in Fig. 6.

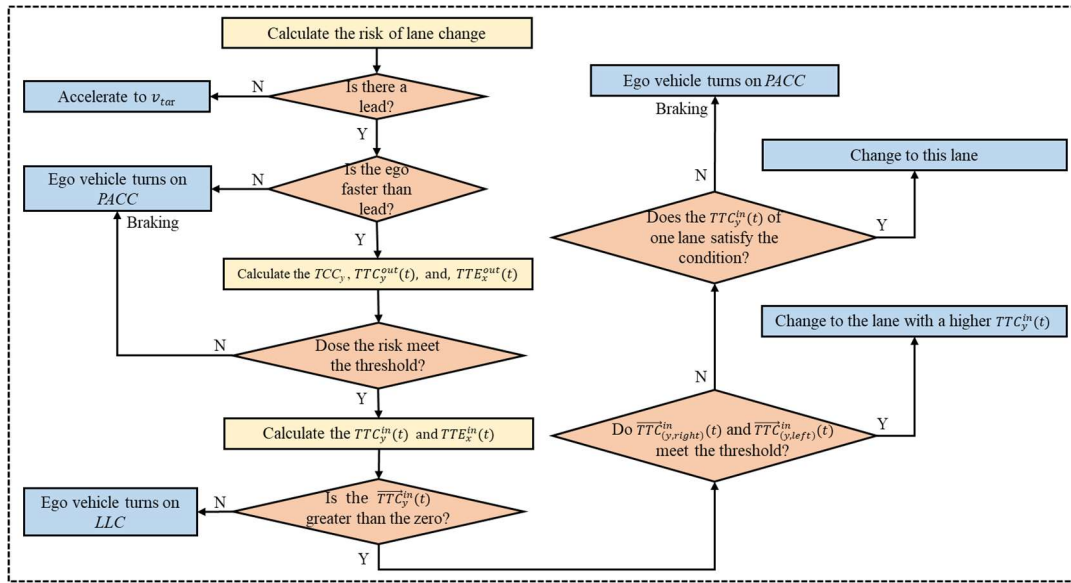


Fig. 6. Decision-making rules and transfer conditions.

4. Personalized vehicle trajectory planning

After completing the decision-making process, the vehicle executes the corresponding maneuvers, which involve free driving, car-following, or lane-changing. However, drivers with different preferences may reach different decisions under the same conditions, or display distinct behaviors even when making the same decision. Such differences are reflected in the timing of lane changes, acceleration patterns, cruising speed, and following distance. To capture these variations, we design a personalized trajectory planning model that accounts for both straight driving and lane-changing scenarios, adapting to individual driving characteristics. In the case of straight driving, the longitudinal trajectory planning is governed by the personalized adaptive cruise control (PACC) system, which dynamically regulates speed, acceleration, and following distance to ensure both safety and comfort. When a lane change is required, the lane-change trajectory planner operates in conjunction with the speed pre-adjustment (SP-A) system to achieve coordinated lateral and longitudinal control, thereby ensuring a smooth and safe transition between lanes.

4.1. Longitudinal PACC for car-following

In car-following scenarios, the vehicle's longitudinal speed must be dynamically adjusted to

ensure safe and comfortable driving. Using onboard sensors, the vehicle measures the relative distance and speed difference with the lead vehicle. Based on this data, an algorithm determines whether safety constraints are met. The vehicle then adjusts its acceleration and speed accordingly to maintain safe following behavior. To further improve traffic efficiency, a personalized dynamic following strategy is developed based on individual driving preferences, enabling customized longitudinal adaptive cruise control within safe limits.

During adaptive cruise control, the ego vehicle must dynamically adjust its speed based on the speed of the lead vehicle. This helps minimize the speed difference between the two vehicles, improving both driving comfort and traffic flow efficiency. Unlike in collision risk assessment, the TTC metric can become ineffective in this scenario. When the ego vehicle and the lead vehicle maintain a similar speed for an extended period, TTC tends to approach infinity regardless of the actual gap between them. This misleadingly indicates no collision risk. However, if the ego vehicle follows too closely without maintaining a safe buffer, there's still a significant risk—especially if the lead vehicle suddenly brakes or experiences a malfunction. To ensure safety during adaptive cruise control, the THW metric is used instead. THW helps maintain a safe following distance between the ego vehicle and the lead vehicle, reducing the risk of rear-end collisions (as shown in Fig. 7).

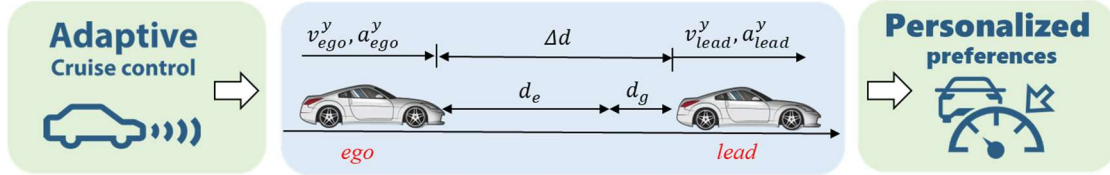


Fig. 7. Vehicle longitudinal PACC module.

Given the relative positions and velocities of the vehicles, their kinematic relationship can be mathematically formulated as follows:

$$\begin{cases} \Delta d = d_g + d_e^{p(A,N,C)} \\ \Delta v = v_{lead}^y - v_{ego}^y \\ d_e^{p(A,N,C)} = T_h^{p(A,N,C)} v_{ego}^y + d_0 \end{cases} \quad (18)$$

$$T_h^{p(A,N,C)} = \frac{y_{lead} - y_{ego}}{v_{ego}^y} \quad (19)$$

where, Δd represents the following distance, d_g is the distance error term, $d_e^{p(A,N,C)}$ denotes the desired following distance based on different driving styles, and d_0 stands for the minimum safe distance. $T_h^{p(A,N,C)}$ refers to the THW, which varies according to different driving preferences. As a result, this leads to different values of $d_e^{p(A,N,C)}$, the desired following distance.

Considering the dynamic limitations of vehicles, acceleration is generally constrained within predefined upper and lower bounds. These constraints ensure safety and comfort by preventing insufficient deceleration that could lead to collisions and by avoiding abrupt acceleration or braking in non-emergency situations. The specific bounds are defined as follows:

$$\begin{cases} v_{min} \leq v_y(t) \leq v_{max} \\ a_{min} \leq a_y(t) \leq a_{max} \\ d_{safe} \leq d(t) \leq d_{max} \end{cases} \quad (20)$$

where, $v_y(t)$ represents the vehicle's speed at time t , v_{min} is the minimum speed limit of the current road, v_{max} is the maximum speed limit of the current road. $a_y(t)$ is the vehicle's acceleration at time t , a_{min} is the vehicle's maximum deceleration, a_{max} is the vehicle's maximum

acceleration. $d(t)$ is the distance between the vehicle and the vehicle ahead or behind at time t , d_{safe} is the minimum safe distance between two vehicles, d_{max} is the vehicle's sensors detection range. In practice, the vehicle's actual acceleration does not respond instantaneously to the desired input but exhibits a time delay. This response behavior is modeled as a first-order inertial system, as shown in Eq. (21).

$$a_y(t) = \frac{K_g}{\tau s + 1} a_e(t) \quad (21)$$

where, $a_y(t)$ represents the vehicle's actual acceleration, $a_e(t)$ is the desired acceleration, τ denotes the inertia (or lag) coefficient, K_g is the system gain, and s is the time step.

4.2. Trajectory planning for lane-changing

In lane-changing scenarios, the vehicle's trajectory is planned using a quintic polynomial (QP) parametric formulation for both longitudinal and lateral motions, integrated with the vehicle's kinematic model. By constraining the trajectory curvature, the steering angle of the front wheels is kept within allowable bounds, thereby satisfying the minimum turning radius requirement. This ensures that the generated lane change trajectories are smooth, comfortable, and efficient. The QP method is adopted because of its ability to produce smooth curves with low computational complexity and high efficiency, making it suitable for real-time applications and adaptable to different driving styles. In this formulation, the vehicle's center is selected as the control point, with x and y denoting its coordinate positions. The equations are expressed as follows:

$$\begin{cases} x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \\ y(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5 \end{cases} \quad (22)$$

where, $x(t)$ and $y(t)$ represent the vehicle's lateral and longitudinal positions at time t , respectively. $a_0, a_1, \dots, a_5$ and $b_0, b_1, \dots, b_5$ are the parameters of the quintic polynomial trajectory. By taking higher-order derivatives of the parametric equations, the vehicle's lateral and longitudinal velocities and accelerations can be calculated. The formulas are as follows:

$$\begin{cases} x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \\ x'(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4 \\ x''(t) = 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3 \end{cases} \Leftrightarrow \begin{cases} y(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5 \\ y'(t) = b_1 + 2b_2 t + 3b_3 t^2 + 4b_4 t^3 + 5b_5 t^4 \\ y''(t) = 2b_2 + 6b_3 t + 12b_4 t^2 + 20b_5 t^3 \end{cases} \quad (23)$$

where, $x'(t)$ and $y'(t)$ represent the vehicle's lateral and longitudinal velocities at time t , respectively, while $x''(t)$ and $y''(t)$ denote the vehicle's lateral and longitudinal accelerations at time t . Given the initial and final positions, velocities, and accelerations of the lane change trajectory, the trajectory equations can be simplified accordingly. By solving for a_0 , a_1 , a_2 , and incorporating the lane change duration parameter $T = \{T | T_A, T_N, T_C\}$, a system of linear equations is established. The solution of this system provides all parameters necessary to construct the complete lane-change trajectory. The detailed formulation is as follows:

$$\begin{bmatrix} T^3 & T^4 & T^5 \\ 3T^2 & 4T^3 & 5T^4 \\ 6T & 12T^2 & 20T^3 \end{bmatrix} \begin{bmatrix} a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} x_{end} - x_{start} - v_{start} T - 0.5a_{start} T^2 \\ v_{end} - v_{start} - a_{start} T \\ a_{end} - a_{start} \end{bmatrix} \quad (24)$$

where, x_{start} , x_{end} , v_{start} , v_{end} , a_{start} , and a_{end} denote the vehicle's initial and final positions, velocities, and accelerations during the lane-change maneuver. By specifying these kinematic parameters, all the coefficients of the longitudinal lane change trajectory can be determined. The same procedure is applied in the lateral direction to obtain the complete trajectory profile.

In this study, safety is imposed as a hard constraint: every trajectory must first satisfy minimum

safety requirements. On this basis, trajectory parameters are adjusted according to individual driving style preferences, ensuring that the final lane-change path is both safe and personalized. As illustrated in Fig. 8, the resulting trajectories quantitatively capture different driving styles. It is clear that trajectory $T = T_C$ reflects a cautious driving style, where the vehicle exhibits lower lateral acceleration and follows a smoother, more gradual curve. This provides a more comfortable driving experience, but at the cost of lower lane change efficiency. In contrast, trajectory $T = T_A$ represents an aggressive driving style, with higher lateral acceleration and a steeper curve, enabling a quicker and more efficient lane change. Additionally, because the cautious style involves lower lateral acceleration, the vehicle tends to travel a longer path longitudinally, giving it more time to respond to traffic changes.

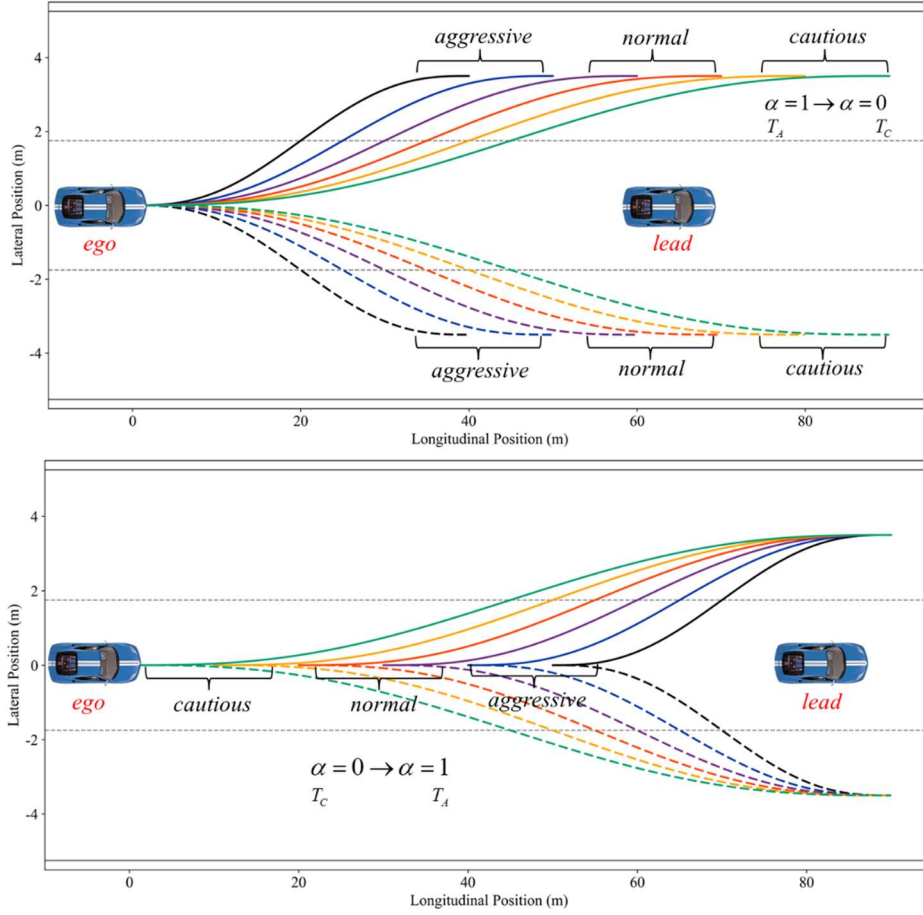


Fig. 8. Lane change trajectory planning considering driving preferences.

During a lane change, effective coordination between the trajectory planning module and the longitudinal PACC system is essential for mitigating potential risk. To further enhance safety and efficiency, a SP-A module is introduced. Prior to initiating a lane change, the SP-A module adjusts the ego vehicle's speed and acceleration according to its current state and surrounding traffic conditions. This adjustment allows the vehicle to better synchronize with the prevailing speed of the target lane, thereby facilitating smoother merging. The SP-A module serves a dual purpose: it contributes to maintaining smooth and efficient traffic flow, while simultaneously reducing the likelihood of unsafe situations during lane changes. In this study, the SP-A module is designed using the fuzzy logic theory. It takes the longitudinal distance difference y_{ea} , normalized speed difference v_{yp} , and lane change risk TTC_y^{in} as inputs. The output is a_{ye} , which determines the pre-adjusted speed for the lane change. The corresponding calculation is given by:

$$y_{ea} = \begin{cases} |y_{ego} - y_{adj}|, adj = 1 \\ 300, adj = 0 \end{cases} \quad (25)$$

where, y_{ea} represents the longitudinal distance difference between the ego vehicle and the adjacent vehicle in the target lane. adj is the state label of the adjacent vehicle, defined as a binary variable (0 or 1). When $adj = 1$, it indicates that an adjacent vehicle is present within the detection range. If no vehicle is detected ($adj = 0$), the distance difference defaults to the maximum detection range of the vehicle's sensors, which is 300 meters.

$$v_{yp} = \frac{|v_{ego}^y - (v_{adj}^y, v_{tar}^y)|}{v_{ego}^y} \quad (26)$$

where, v_{yp} is the normalized value of the speed difference between the ego vehicle and the adjacent vehicle in the target lane. v_{adj}^y and v_{tar}^y represent the longitudinal speed of the adjacent vehicle and the desired longitudinal speed for the target lane, respectively, which varies depending on the driving style. If there is an adjacent vehicle in the target lane, the speed difference is calculated between the ego vehicle and that vehicle. Otherwise, the difference is computed between the ego vehicle's speed and the v_{tar}^y . The calculation method for the lane change risk TTC_y^{in} has already been presented in the risk assessment section.

(1) The longitudinal distance y_{ea} between the lane-changing vehicle and the adjacent vehicle in the target lane plays a critical role in ensuring lane-change safety and in adjusting speed in advance. To maintain smoothness and robustness in SP-A, three gradually transitioning membership functions are used to represent this parameter. The membership function is defined based on the d_r , which corresponds to the distance the ego vehicle travels during the driver's perception and braking response time. The values of y_{ea} are classified into three levels—*high*, *medium*, and *low*—based on the range of d_r . A higher level indicates a larger distance between the ego vehicle and adjacent vehicles, corresponding to lower potential collision risk. The membership function for y_{ea} is as follows:

$$y_{ea} = \begin{cases} low, y_{ea} \in (0, d_r^l] \\ mid, y_{ea} \in (d_r^l, d_r^m] \\ high, y_{ea} \in (d_r^m, d_r^h] \end{cases} \quad (27)$$

(2) The normalized speed difference v_{yp} between the lane-changing vehicle and the vehicle in the adjacent lane represents the proportion of their speed difference. Based on the range of the normalized speed difference between the two vehicles, v_{yp} can also be classified into three levels: *high*, *medium*, and *low*. A higher level indicates a larger speed gap between the ego vehicle and the target-lane vehicle, implying a greater need for speed pre-adjustment. The membership function for v_{yp} is as follows:

$$v_{yp} = \begin{cases} low, v_{yp} \in (0, v_{yp}^l] \\ mid, v_{yp} \in (v_{yp}^l, v_{yp}^m] \\ high, v_{yp} \in (v_{yp}^m, v_{yp}^h] \end{cases} \quad (28)$$

(3) The collision risk TTC_y^{in} between the lane-changing vehicle and the vehicle in the adjacent lane is classified into three membership levels based on the severity of the risk. A smaller TTC_y^{in} indicates a higher collision risk, in which case risk avoidance becomes the top priority and the need for speed pre-adjustment correspondingly decreases. The membership function for TTC_y^{in} is defined as follows:

$$TTC_y^{in} = \begin{cases} low, TTC_y^{in} \in (0, TTC_y^{(in,l)}] \\ mid, TTC_y^{in} \in (TTC_y^{(in,l)}, TTC_y^{(in,m)}] \\ high, TTC_y^{in} \in (TTC_y^{(in,m)}, TTC_y^{(in,h)}] \end{cases} \quad (29)$$

The lane change SP-A module employs continuous traffic operation parameters (e.g., y_{ea} , v_{yp} , and TTC_y^{in}) as inputs to a fuzzy logic controller that regulates the vehicle's acceleration. The output is the vehicle's acceleration state $a_{ye} = (acc, null, bra)$, which can take one of three actions: acceleration, cruising, or braking. Based on this process, the SP-A strategy for the lane-changing vehicle is generated (as shown in Table 1). When the normalized speed difference between the ego vehicle and the target-lane vehicle is larger, the relative distance is greater, and TTC is higher (i.e., safer), the SP-A output of the ego vehicle tends to favor acceleration. However, when safety conditions are not met, SP-A tends to output braking even if the speed difference is large.

Table 1 SP-A module parameters for lane-changing.

y_{ea}	v_{yp}	TTC_y^{in}	a_{ye}
--	high	high	acc
high	mid	high	acc
high	high	mid	acc
high	mid	mid	acc
high	low	mid	null
--	low	high	null
low	high	mid	null
low	mid	high	acc
low	mid	mid	null
low	low	mid	bra
--	--	low	bra
mid	high	mid	acc
mid	mid	high	acc
mid	mid	mid	acc
mid	low	mid	null

Note: '--' denotes that any state is acceptable.

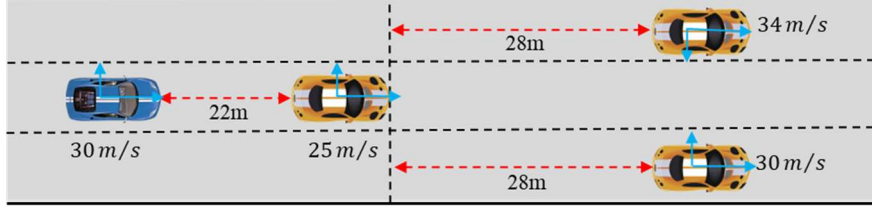
5. Experiment

5.1. Scenarios setting

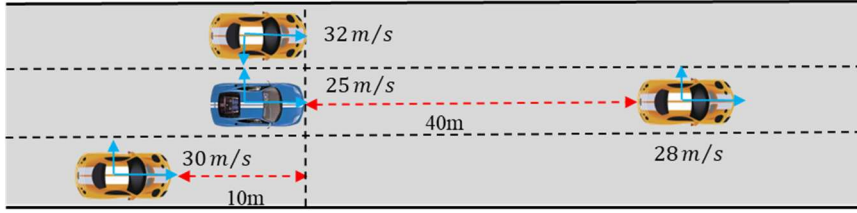
To validate the proposed method, simulation experiments are conducted on a three-lane highway using the HighD dataset. The framework, implemented in Python 3.11, integrates risk assessment, decision-making, and trajectory planning modules. Vehicle and lane parameters are defined based on realistic settings, and representative scenarios are designed to cover free driving, car-following, and lane-changing conditions. Specifically, three types of risk combinations are considered: (1) In Scenario 1, the current lane poses a collision risk, while both the left and right lanes have relatively low risk. (2) In Scenario 2, the current lane is low-risk, but both adjacent lanes have collision risks. (3) In Scenario 3, the current and both adjacent lanes all present collision risks. The scenario parameters are taken as in Table 2 and the scenario setting is illustrated in Fig. 9.

Table 2 Parameter settings in different scenarios.

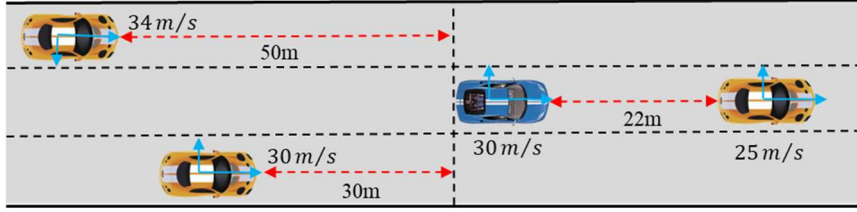
	Ego vehicle		Lead vehicle		Right vehicle		Left vehicle	
	Position	Velocity	Position	Velocity	Position	Velocity	Position	Velocity
S.1	(0,4.25)	30m/s	(22, 4.25)	25m/s	(50,1.75)	30m/s	(7.75,50)	34m/s
S.2	(10, 4.25)	25m/s	(50, 4.25)	28m/s	(0, 1.75)	30m/s	(10, 7.75)	32m/s
S.3	(50, 4.25)	30m/s	(72, 4.25)	25m/s	(20,1.75)	34m/s	(0, 7.75)	34m/s



a. Scenario 1



b. Scenario 2



c. Scenario 3

Fig. 9. Simulation scenarios setting.

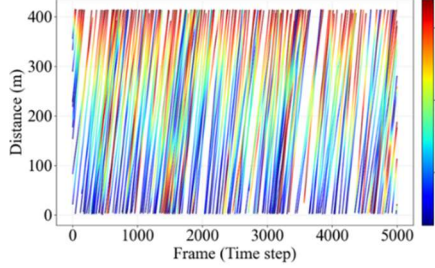
5.2. Results and discussion

5.2.1 Driving preference analysis

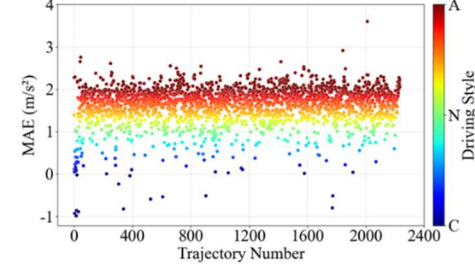
To comprehensively characterize driving preferences across different styles, we analyze five key indicators: MAE, THW, DHW, TTC, and LCD. The analysis draws on large-scale naturalistic driving data from the HighD dataset. First, the GMM clustering model incorporating traffic knowledge is applied to classify driving styles. As shown in Fig. 10, the styles are categorized into aggressive, normal, and cautious. A higher MAE corresponds to a more aggressive style, while cautious drivers exhibit acceleration patterns closer to those predicted by the IDM. Since these two parameters are correlated to some extent, analyzing them jointly helps reveal their complex relationship and supports more informed and rational parameter design for the personalized driving system. To further analyze behavioral features, a dual-dimensional interaction analysis of THW and DHW is performed (as shown in Fig. 11). The color density reflects trajectory frequency, with purple indicating low and red indicating high concentrations. The $H_proportion$ highlights the typical THW–DHW distribution for each style. Cautious drivers maintain larger gaps and reaction times, prioritizing safety, whereas aggressive drivers adopt smaller values, improving efficiency while relying on skill to ensure safety.



a. High_D natural driving trajectory



b. Distance - velocity distribution



c. Driving style - MAE distribution

Fig. 10. Sample analysis of natural driving trajectory data.

Fig. 12 illustrates the TTC distributions for the three driving styles. Cautious drivers show the smallest blue area, indicating larger average TTC values and lower risk. Aggressive drivers display a wider blue area with smaller TTC values, reflecting higher risk tolerance and more assertive behavior. Normal drivers fall between these two extremes, balancing safety and efficiency. These profiles provide important guidance for tuning decision-making and trajectory planning parameters in the personalized autonomous driving framework.

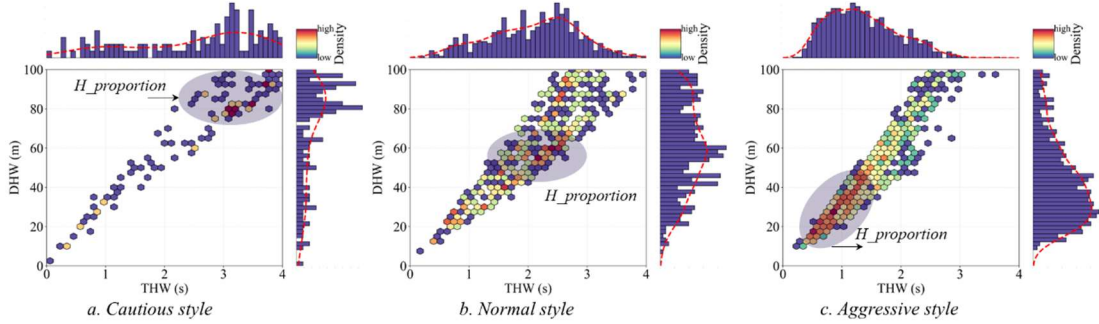


Fig. 11. THW and DHW characterization for different driving styles.

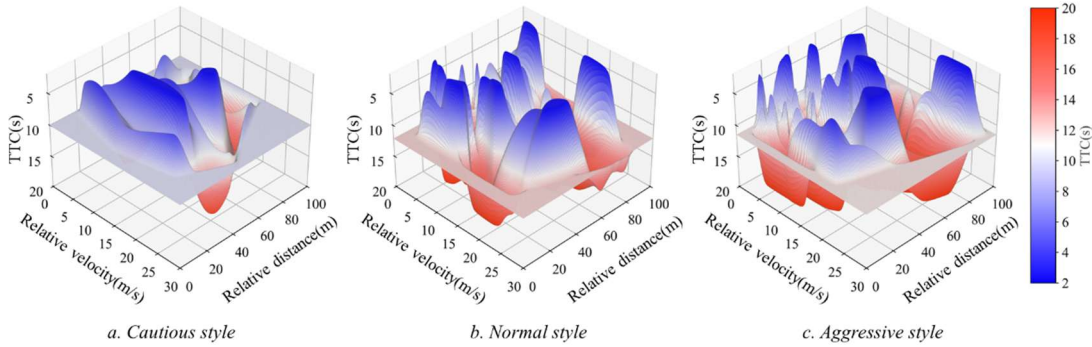


Fig. 12. Driving risk characterization of different driving styles.

To support personalized trajectory planning, we analyze lane change durations (LCDs), focusing on the differences across driving styles. Lane change events are extracted from naturalistic trajectories, with start and end frames identified through slope-based detection and manual verification. As shown in Fig. 13, the full lane change trajectory and acceleration profile for a sample vehicle are visualized, with the start and end frames clearly marked. This lane change lasts 160

frames, corresponding to an LCD of 6.4 seconds. The lateral displacement is -3.5 m, approximately equal to the width of a single lane. Both the displacement and lateral acceleration are negative, indicating that the vehicle shifts from the inner lane to the outer lane. This trajectory closely resembles a quintic polynomial curve, confirming the appropriateness of the trajectory planning model adopted in this study. Across the entire dataset, LCDs range from 3–12 s, with most between 3–7 s (as shown in Fig. 14a). Style-specific analysis shows cautious drivers prefer longer LCDs (6–8 s), normal drivers fall between 4–6 s, and aggressive drivers concentrate between 3–5 s (as shown in Fig. 14b–Fig. 14d).

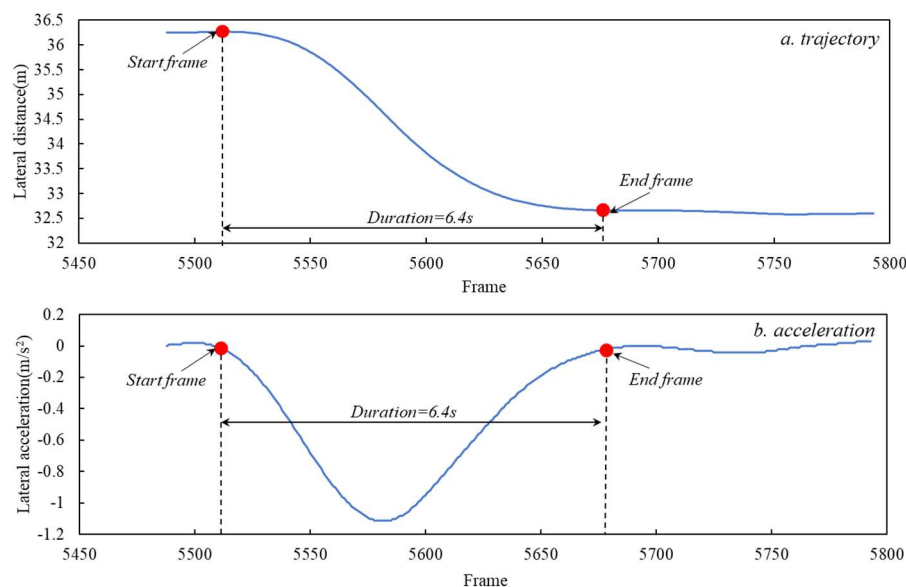


Fig. 13. Lane changing process for a sample vehicle.

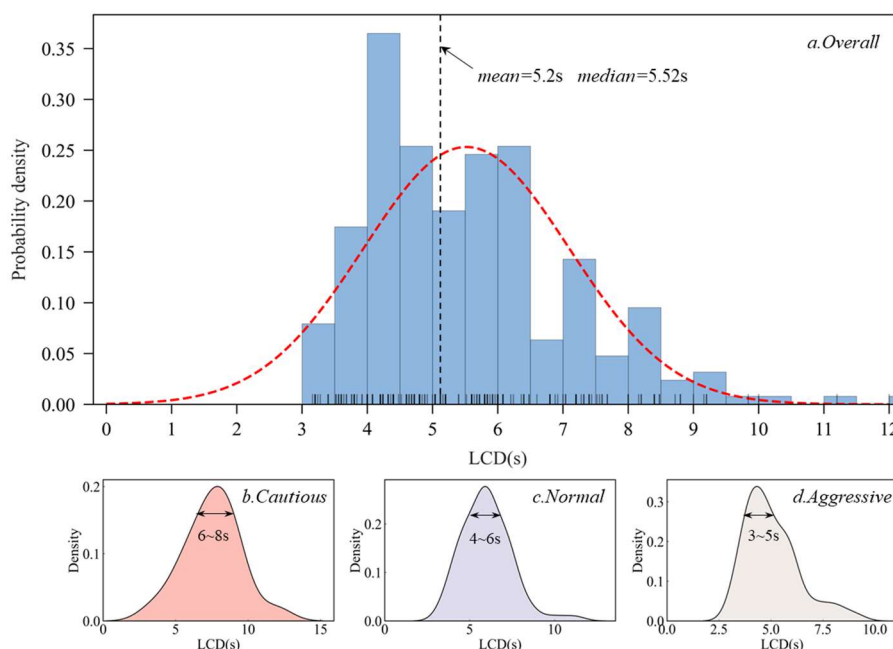


Fig. 14. Probability density distribution for the distribution of LCDs.

5.2.2 Simulation analysis

I. Scenario 1: The middle lane is risky

In Scenario 1, the ego vehicle faces a high longitudinal collision risk with the lead vehicle, while

the left and right lanes present minimal risk. The personalized decision-making and trajectory planning modules are applied, resulting in distinct behaviors for different driving styles. Fig. 15 visualizes the driving risk and longitudinal trajectories using normalized heatmaps. At the current time, the middle lane exhibits the highest collision risk, whereas the adjacent lanes are safer. Consequently, the ego vehicle initiates a lane change under all driving styles. The longitudinal position curves indicate that the cautious style completes the overtaking maneuver last, the normal style earlier, and the aggressive style earliest. Correspondingly, the aggressive trajectory exhibits the steepest slope and highest longitudinal speed, followed by the normal style, while the cautious style maintains the lowest speed.

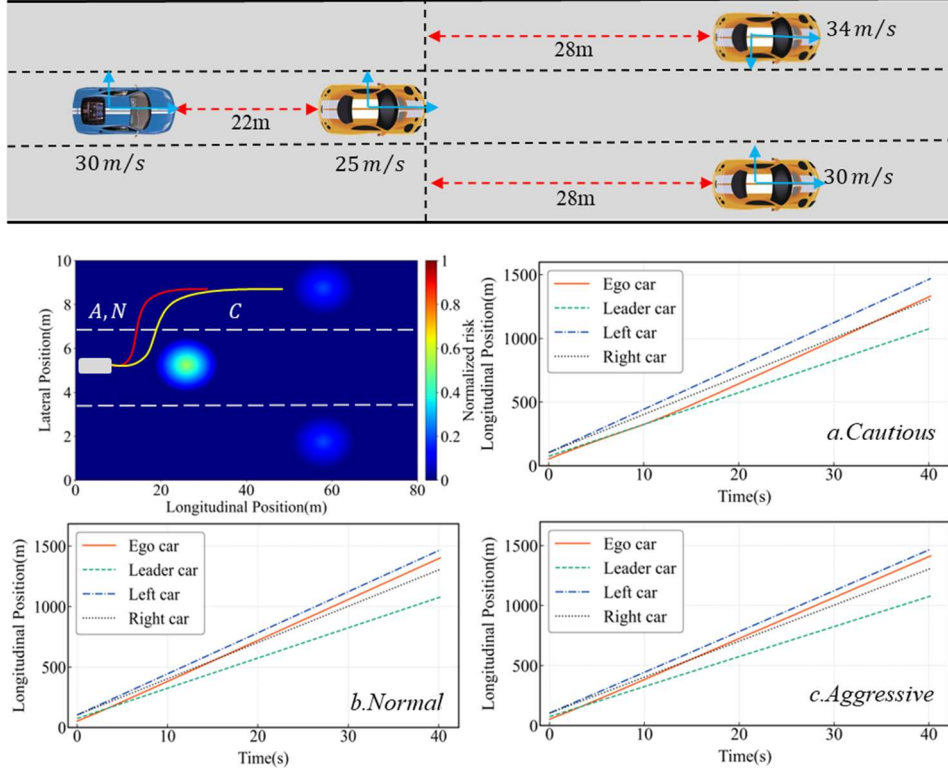
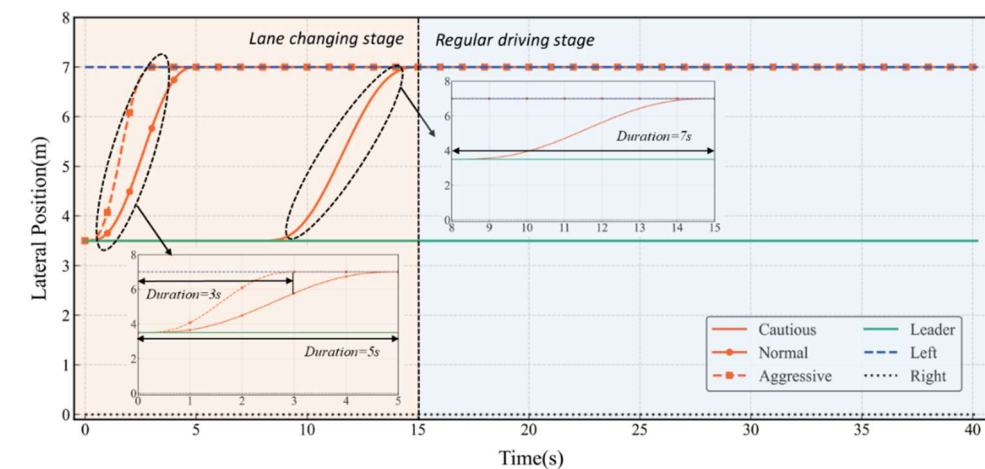


Fig. 15. Driving risk and longitudinal position in scenario 1.

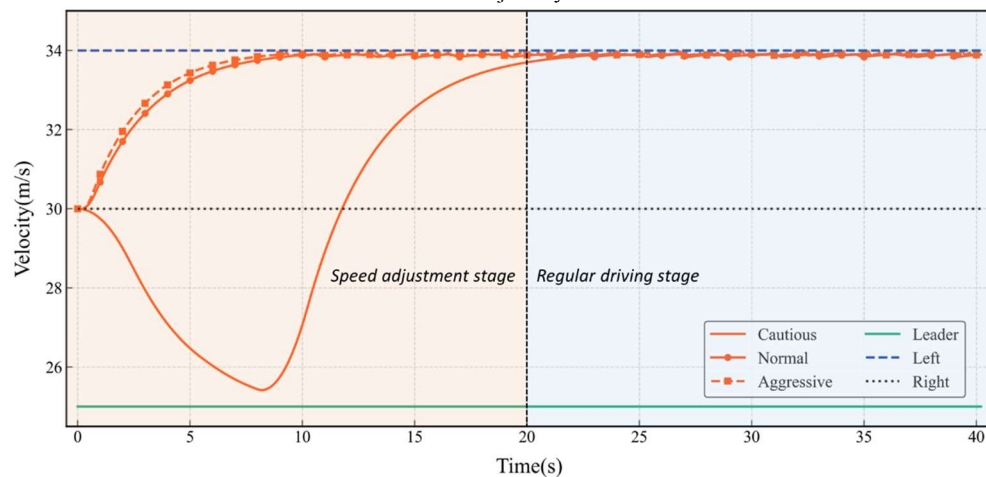
Fig. 16a shows the lateral trajectories. The vertical axis marks lane centerlines, with “0” for the centerline of the rightmost lane and “7” for the leftmost. Aggressive and normal styles initiate direct lane changes, while the cautious style first decelerates to follow the lead vehicle before changing lanes around 8 seconds. All lane changes follow QP curves, producing smooth trajectories. The LCDs differ by style: aggressive completes around 3 s, normal around 5 s, and cautious around 7 s, reflecting their respective risk preferences. Further analysis of the vehicle speeds in Scenario 1 is shown in Fig. 16b. Aggressive and normal styles accelerate during the maneuver and then use PACC to follow the front-left vehicle, whereas the cautious style decelerates first to maintain safety and only accelerates after leaving the current lane. The rate of acceleration varies by style, with aggressive drivers accelerating fastest and cautious drivers more gradually.

Fig. 16c illustrates acceleration patterns. Aggressive and normal styles maintain positive acceleration from 0 to 10 seconds, performing accelerated lane changes, with more rapid fluctuations in the aggressive style. The cautious style shows negative acceleration from 0 to 8 seconds to mitigate collision risk, followed by gradual acceleration between 8 and 20 seconds to merge safely and align with traffic in the target lane. Moreover, the *acceleration adjustment stage* is longer for the cautious style. The cautious style requires a longer lane-changing time window,

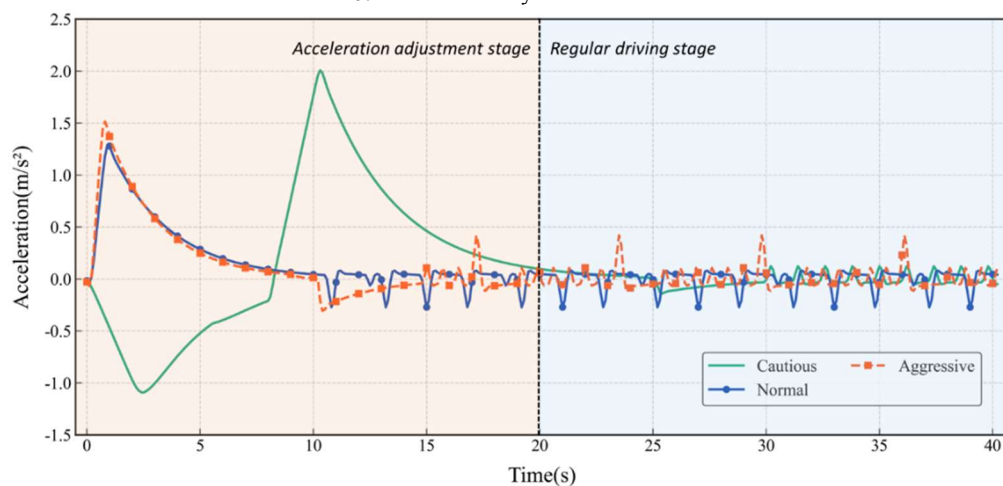
whereas aggressive and normal styles complete adjustments more quickly. During *regular driving stage*, all styles maintain safe and comfortable acceleration profiles in the target lane. Fluctuations are most pronounced in the aggressive style and minimal in the cautious style, but remain within human comfort limits.



a. Vehicle lateral trajectory in scenario 1.



b. Vehicle velocity in scenario 1.



c. Vehicle acceleration in scenario 1.

Fig. 16. Lateral trajectory, velocity, and acceleration of vehicles in scenario 1.

II. Scenario 2: The lanes on both sides are risky

In Scenario 2, the ego vehicle faces minimal collision risk in the current lane but higher risks in

both adjacent lanes. As shown in Fig. 17, the left lane poses the greatest risk, followed by the right, while the current lane is the safest. Under cautious and normal driving styles, the ego vehicle remains in the center lane, following the lead vehicle throughout. By contrast, the aggressive style initially follows the lead vehicle but executes a left lane change between 6–7 seconds once conditions improve, successfully overtaking in the longitudinal direction.

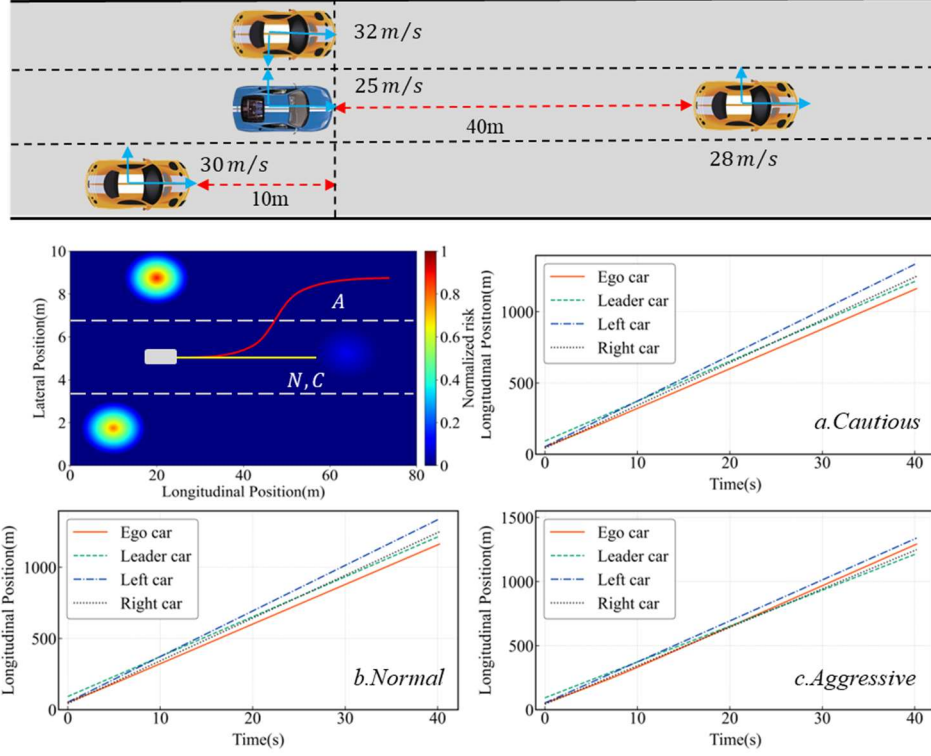
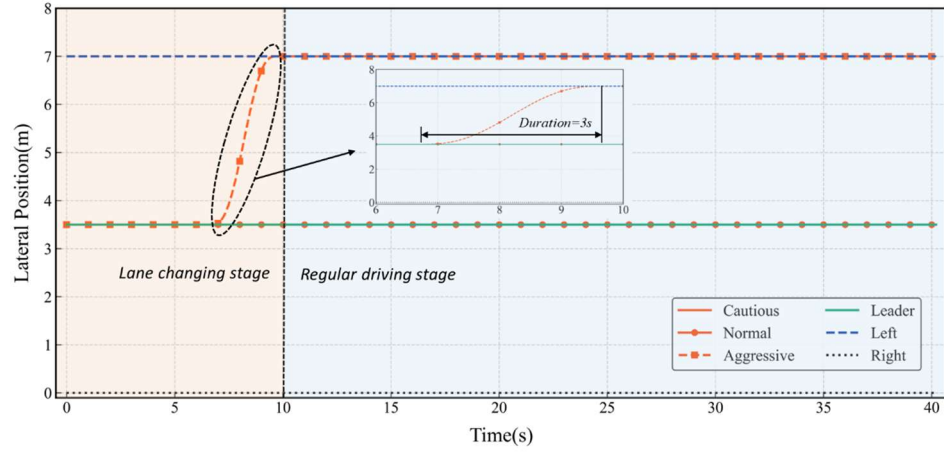


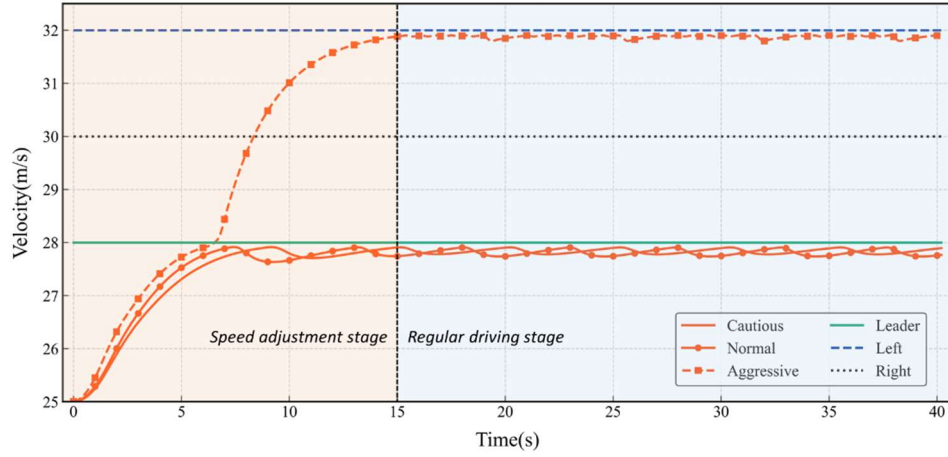
Fig. 17. Driving risk and longitudinal position in scenario 2.

The lateral trajectories are illustrated in Fig. 18a. During the first 6 seconds, none of the styles attempt a lane change due to risks in adjacent lanes. As the left lane becomes safer, the aggressive style initiates a smooth QP lane change lasting around 3 seconds. Speed profiles (as shown in Fig. 18b) show that all styles adapt acceleration via the proposed PACC to improve traffic flow, but with different rates: aggressive drivers accelerate fastest, cautious drivers slowest, and normal drivers in between. Since both adjacent lanes present relatively high collision risks, vehicles under cautious and normal styles remain in the center lane, following the lead vehicle via PACC. In contrast, the aggressive style, characterized by greater risk tolerance, accelerates again and initiates a left lane change once safety conditions permit. After completing the maneuver, the aggressive style re-engages PACC, accelerates to 32 m/s, and follows the preceding vehicle in the target lane.

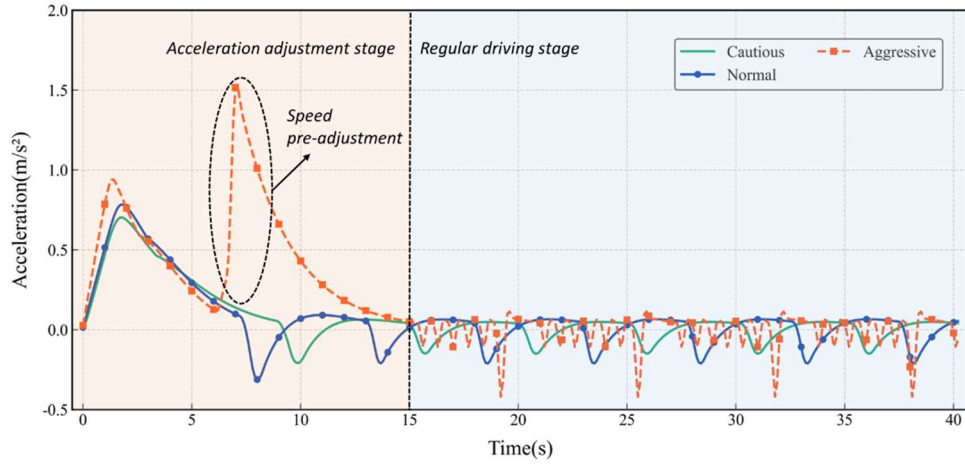
Acceleration patterns (as shown in Fig. 18c) further highlight style-dependent behaviors. In the *acceleration adjustment stage*, all driving styles exhibit acceleration. However, the magnitudes and durations differ: the aggressive style reaches the target speed within 6 seconds, the normal style within 7 seconds, and the cautious style within 9 seconds. The aggressive profile shows the largest and fastest variations, while the cautious profile is smoother and more gradual. In the *regular driving stage*, all styles follow the lead vehicle, though the aggressive style achieves higher efficiency. Across scenarios, accelerations remain within $\pm 2.5 \text{ m/s}^2$, ensuring ride comfort.



a. Vehicle lateral trajectory in scenario 2.



b. Vehicle velocity in scenario 2.



c. Vehicle acceleration in scenario 2.

Fig. 18. Lateral trajectory, velocity, and acceleration of vehicles in scenario 2.

III. Scenario 3: All lanes are risky

In Scenario 3, the ego vehicle faces collision risks from the lead vehicle in the same lane and from vehicles in both adjacent lanes. As shown in Fig. 19, all three lanes present risks, with the center lane being the most hazardous. Initially, the ego vehicle is ahead of the left and right vehicles but constrained by the lead vehicle. Under the cautious style, the ego vehicle avoids immediate lane changing, instead decelerating and waiting for the adjacent vehicles to pass before initiating a maneuver. The normal style chooses to change lanes earlier, before being overtaken, and then

overtakes the lead vehicle after merging. The aggressive style also changes lanes immediately but at a higher speed, steadily increasing its distance from surrounding vehicles.

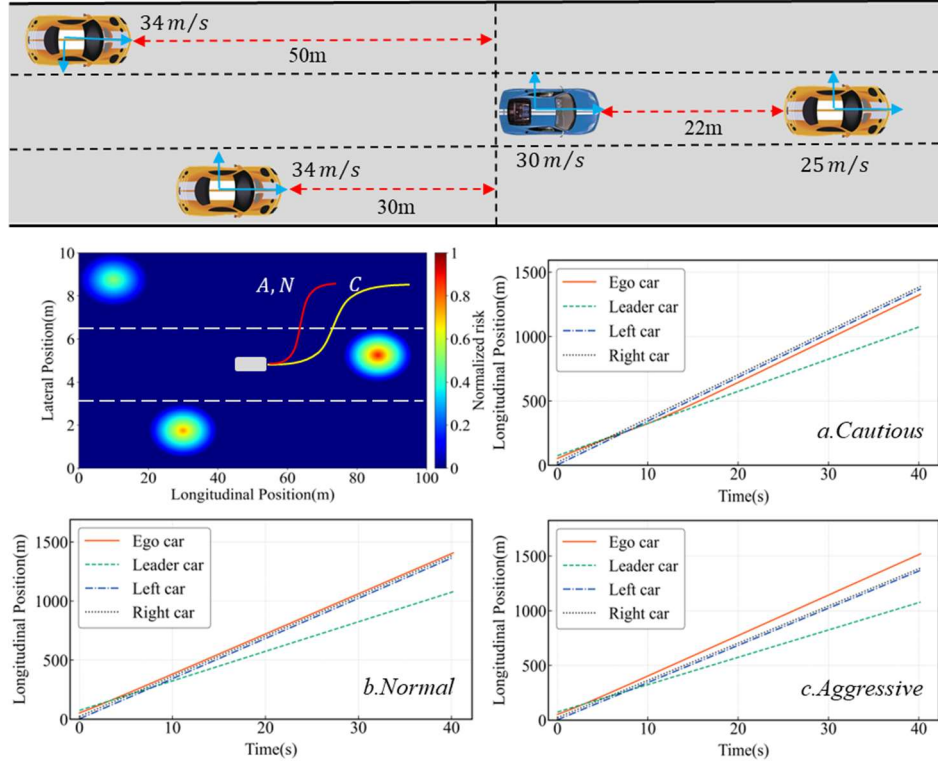
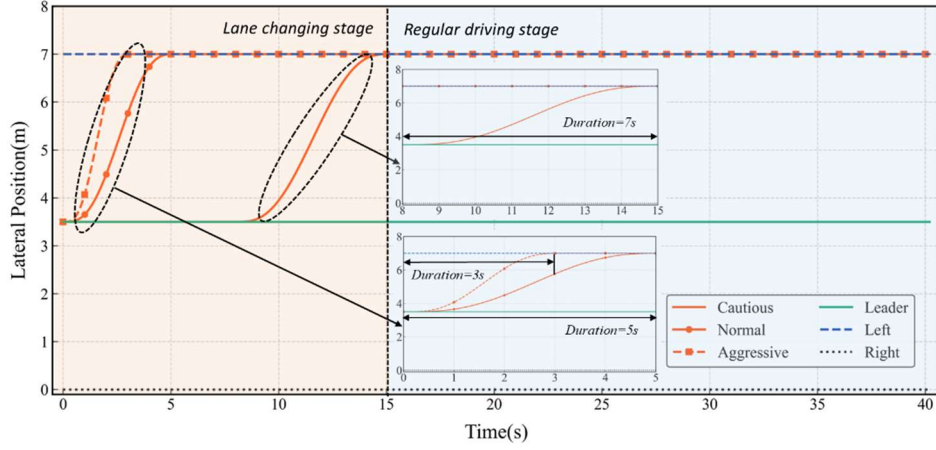


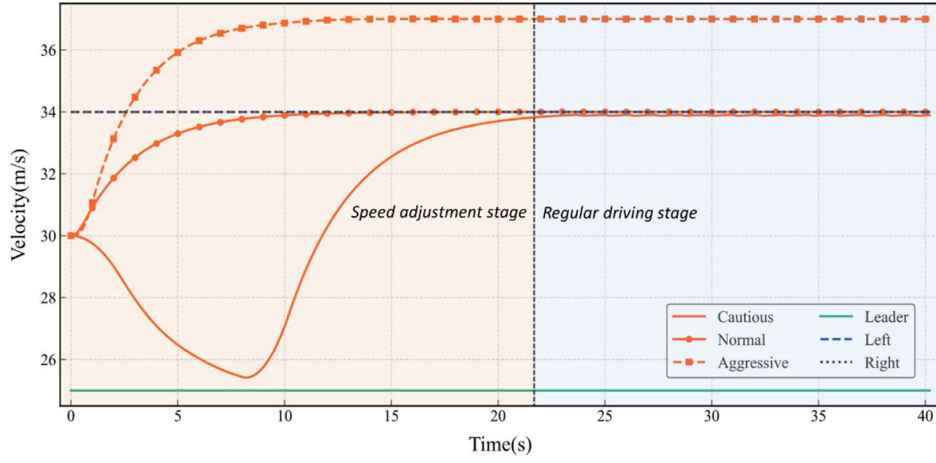
Fig. 19. Driving risk and longitudinal position in scenario 3.

The lateral trajectories (as shown in Fig. 20a) reveal that both the aggressive and normal styles change lanes to the left directly, completing smooth QP-based trajectories in about 3–5 s. The cautious style, however, first decelerates to follow the lead vehicle and only initiates a lane change at around 8 s, taking about 7 s to complete the maneuver with the smoothest and gentlest trajectory. Speed profiles (as shown in Fig. 20b) further highlight style-dependent strategies. Aggressive and normal styles accelerate during the maneuver and reach cruising speeds of 37 m/s and 34 m/s, respectively. In contrast, the cautious style first decelerates to reduce collision risk, allowing adjacent vehicles to pass, and then accelerates via SP-A during the lane change to adapt to target-lane traffic. After merging, all styles engage PACC, though the cautious style follows a preceding vehicle rather than achieving free-flow.

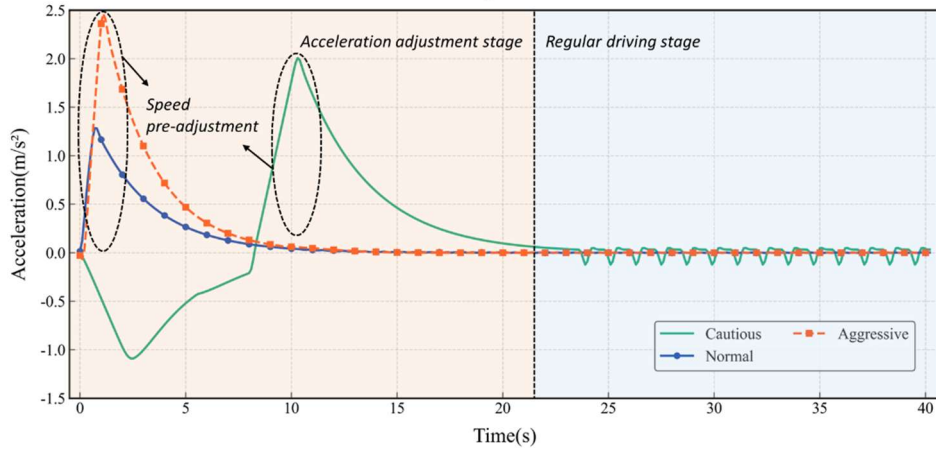
Acceleration curves (as shown in Fig. 20c) confirm these distinctions. In the *acceleration adjustment stage*, the aggressive style shows the highest peak acceleration, followed by the normal style, while the cautious style exhibits negative acceleration in the first 8 s before switching to positive acceleration. This reflects the initial deceleration to mitigate collision risk, and a subsequent acceleration to merge safely into the target lane. In the *regular driving stage*, aggressive and normal styles sustain constant speeds in free flow, whereas the cautious style follows the new lead vehicle. Across all cases, acceleration remains within human comfort thresholds. Overall, this scenario illustrates how the framework accommodates diverse driving preferences: aggressive drivers prioritize efficiency, normal drivers balance safety and performance, and cautious drivers adopt conservative but safer strategies.



a. Vehicle lateral trajectory in scenario 3.



b. Vehicle velocity in scenario 3.



c. Vehicle acceleration in scenario 3.

Fig. 20. Lateral trajectory, velocity, and acceleration of vehicles in scenario 3.

5.3. Comparison analysis

The personalized autonomous driving system developed in this study ensures safe, comfortable, and user-tailored vehicle operation in complex three-lane scenarios. In environments with multiple vehicles and coupled collision risks, the system supports essential maneuvers, including lane keeping, car-following, and lane changing. It integrates the PACC for regular driving phases and the SP-A module for lane-changing phases, both of which enhance driving safety and efficiency.

To further evaluate performance, we analyze all risk-combination scenarios through comparative

experiments. Efficiency is assessed by measuring the time required for vehicles to traverse each scenario, from entry to exit. Because travel distance may vary across different scenarios, we apply a fixed threshold of 1000 meters, within which vehicles typically complete all maneuvers and transition to cruise control in multiple preliminary experiments. As shown in Fig. 21, dots represent maneuver completion times, while the blue shaded areas indicate the time saved by the personalized strategy compared with a single-style baseline. *Traditional single* serves as the baseline, representing a conventional autonomous driving decision-making and planning model based on a single driving style. *Personalized-N* refers to the personalized decision-making and planning model proposed in this study operating under a normal driving style, while *Personalized-A* denotes the same personalized model operating under an aggressive driving style. Figs. 21a and 21b illustrate completion times and relative time savings for both lane-change maneuvers and full-scenario traversal. Across all three risk scenarios, the personalized system achieves higher traffic efficiency. The advantage is particularly evident in the high-risk Scenario 3, where the aggressive style improves efficiency by 16.6% and the normal style by nearly 10%. As driving distance and task complexity increase, the efficiency gains of the personalized strategy become even more pronounced.

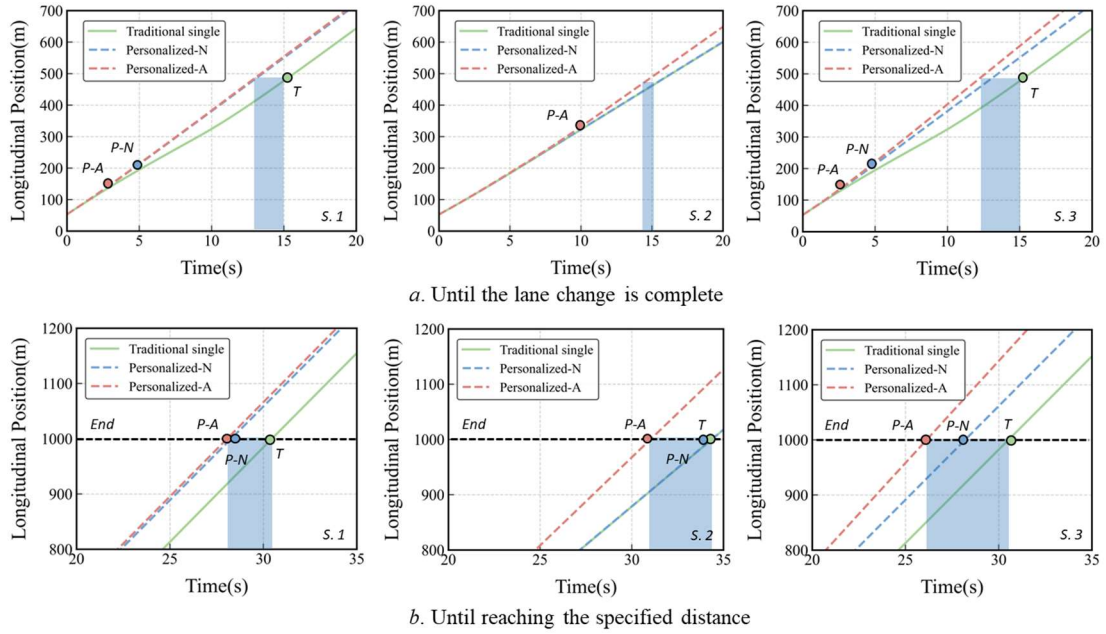


Fig. 21. Overall traffic efficiency of personalized driving systems.

To build on the previous analysis of efficiency gains over the entire driving process, we now focus specifically on efficiency during the lane-changing phase. To ensure fair comparison, only cases where lane changes occur under all driving styles are considered. Efficiency is evaluated from three perspectives: lane change completion, speed adjustment, and acceleration adjustment (as shown in Fig. 22). The results show that personalized strategies consistently outperform the traditional single-style approach. Compared to the baseline, the aggressive style reduces the time required for lane changes, speed adjustment, and acceleration adjustment by 80%, 55%, and 55%, respectively; the normal style achieves reductions of 66%, 50%, and 50%. On average, personalized strategies shorten these times by 48%, 35%, and 35%. These findings indicate that personalization not only accommodates diverse driving preferences but also improves overall efficiency.

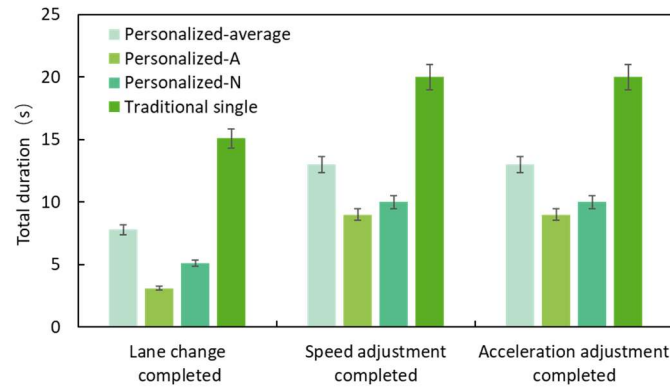


Fig. 22. Comparison of the efficiency of multiple solutions throughout the lane change process.

6. Conclusion

This paper presents a holistic personalized autonomous driving framework that aligns vehicle behavior with driver-specific preferences by integrating driver profiling, dual-dimensional risk assessment, preference-aware decision-making, and individual trajectory planning. Specifically, we quantify deviations between naturalistic driving and IDM predictions, and classify the driving style into three clusters via the GMM. The resulting driving style parameters are input into the downstream pipeline. This enables vehicles to update driving strategies based on historical data and user preferences. A lateral-longitudinal risk model evaluates real-time collision exposure (including cut-in and cut-out phases), and its outputs are integrated with driver preferences in an FSM-based decision layer to generate maneuver-level strategies. A dual-component trajectory planner then converts these strategies into executable trajectories: a PACC for longitudinal control and a fuzzy-logic quintic-polynomial lane-change planner augmented by a SP-A mechanism for smooth, preference-aware merging. Extensive multi-vehicle simulations on three-lane highways demonstrate that the proposed framework reliably captures distinct driving styles, produces accurate risk estimates, and yields safe, comfortable, and user-tailored trajectories. Compared with a conventional single-style baseline, the proposed approach substantially improves operational efficiency (e.g., marked reductions in lane-change durations and faster post-merge cruising), while PACC and SP-A further enhance throughput without degrading safety or personalization. By bridging naturalistic driving data with an autonomous driving framework, this work shows that personalization is both feasible and beneficial. Future work will focus on multi-vehicle coordination, real-world deployment and validation, and online adaptation to further improve scalability and robustness.

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